

PINNs for High-Frequency Helmholtz Problems

Progress Report

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Goal

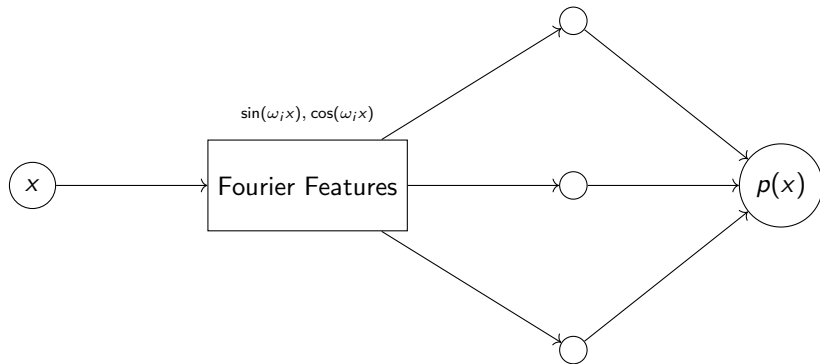
Solve the Helmholtz equation arising in acoustics

$$\nabla^2 p + k^2 p = 0$$

Challenges

- Large wave number $k \rightarrow$ highly oscillatory solutions
- Standard PINNs struggle due to **spectral bias**
- Optimization becomes difficult as frequency increases

Baseline PINN Architecture



- Fully connected network
- Fourier feature embedding
- PDE residual used as loss

Improvement Attempt 1: Larger Fourier Span

Idea:

- Increase frequency range in Fourier features
- Allow network to access higher frequency basis

$$\gamma(x) = [\sin(\omega_1 x), \dots, \sin(\omega_n x)]$$

Observation:

- No significant improvement
- Model concentrates energy around a dominant spectral band

Improvement Attempt 2: Learnable Frequencies

Idea

- Make Fourier frequencies trainable
- Allow network to adapt spectral representation

Observation

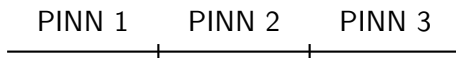
- Optimization becomes unstable
- No consistent improvement in convergence

Key Insight

The network is failing to learn higher frequencies due to the complexity of the loss landscape.

Solving for XPINNs

Domain decomposition



Each subdomain

- Independent PINN
- The continuity of the interface is hardly constrained.
- Soft boundary enforcement over equation of continuity

Advantages with fourier feature model

- Improved convergence
- Better representation of oscillatory solutions
- Reduced optimization instability

Interpretation

XPINNs reduce effective frequency per subdomain

Motivation for Multiplicative Features

Trigonometric identity

$$\sin(x) \cos(y) = \frac{1}{2}[\sin(x + y) + \sin(x - y)]$$

Implication

- Multiplication generates higher harmonics
- Standard networks rely on additive combinations

Idea

Introduce multiplicative interactions

Log-space Multiplicative Representation

Explored idea

$$\prod_j \phi_j(x) = \exp \left(\sum_j \log \phi_j(x) \right)$$

Observed behavior

- Extremely non-convex loss landscape
- Large curvature variations
- Not even obtaining a convergence for the low frequency solution.

Conclusion

Expressive representation but unstable optimization

Any improvements that can be suggested to this model?

Further ideas

- Any regularisation/change in loss function that could help the model learn higher frequencies much better?
- Solving a complex problem using fourier feature model to solve a complex.
- Dynamic weighting, residual-based adaptive resampling (RAR), curriculum learning, and Sobolev training